Python Programming Assessment

Section A: 20 multiple-choice questions

Section B: 10 coding questions

Instructions: Select one answer for each MCQ. Write and test Python programs for Section B.

Section A – Multiple Choice Questions

Q1. What will be the output of the following code?

```
x = 5
if x > 3:
    if x < 5:
        print('A')
    else:
        print('B')
else:
    print('C')
```

A. A
B. B
C. C
D. No output

Answer: _____

Q2. What will be the final value of count?

```
count = 0
for i in range(1, 6):
    if i % 2 == 0:
        continue
    count += i
print(count)
```

A. 6
B. 9
C. 15
D. 3

Answer: _____

Q3. What will the following while loop print?

```
x = 1
while x < 10:
    x *= 2
print(x)
```

A. 8
B. 10
C. 16
D. Infinite loop

Answer: _____

Q4. What is the output?

```
for i in range(3):
    for j in range(2):
        if i == j:
            print(i, j)
```

A. 0 0 and 1 1
B. 0 0, 1 1 and 2 2
C. 0 1 and 1 0
D. No output

Answer: _____

Q5. What is the output of this function?

```
def calculate(a, b=5):
    return a * b
print(calculate(3))
print(calculate(3, 2))
```

A. 15 and 6
B. 15 and 15
C. Error
D. 3 and 6

Answer: _____

Q6. What is printed?

```
def update(value):
    value += 10
    return value
x = 5
update(x)
print(x)
```

A. 15
B. 5
C. 10
D. None

Answer: _____

Q7. What happens when this program runs?

```
def check(n):
    if n > 0:
        return 'Positive'
    print('Finished')
print(check(-2))
```

A. Positive
B. Finished followed by None
C. Finished followed by Positive
D. Error

Answer: _____

Q8. Assume utils.py contains greet(). What is the correct way to call it after importing the module?

```
# utils.py
def greet(name):
    return 'Hello ' + name
# main.py
import utils
```

A. greet('Ravi')
B. utils.greet('Ravi')
C. import.greet('Ravi')
D. utils->greet('Ravi')

Answer: _____

Q9. Which statement correctly imports calculate() from this nested package?

```
project/
    main.py
    tools/
        __init__.py
        mathops/
            __init__.py
            arithmetic.py  # contains calculate()
```

A. import tools.calculate
B. from tools.mathops.arithmetic import calculate
C. from arithmetic import tools
D. import mathops.calculate

Answer: _____

Q10. What is the main reason to use separate virtual environments for two Python projects?

A. Each environment runs on a separate CPU
B. Different projects can use different dependency versions
C. Virtual environments automatically improve execution speed
D. Python requires one virtual environment for every module

Answer: _____

Q11. What does the following command primarily do?

```
python -m pip freeze > requirements.txt
```

A. Installs all dependencies
B. Writes installed package versions to a file
C. Creates a virtual environment
D. Removes unused packages

Answer: _____

Q12. What is the output?

```
def test(*args):
    return args[0] + args[-1]
print(test(10, 20, 30, 40))
```

A. 100
B. 50
C. 30
D. Error

Answer: _____

Q13. What is printed?

```
def display(**kwargs):
    return kwargs.get('age', 0)
print(display(name='Anil', city='Delhi'))
```

A. None
B. KeyError
C. 0
D. Delhi

Answer: _____

Q14. What is the output?

```
def sample(a, *args, **kwargs):
    print(a, len(args), len(kwargs))
sample(1, 2, 3, x=4, y=5)
```

A. 1 2 2
B. 1 3 2
C. 1 2 3
D. Error

Answer: _____

Q15. What will the following code print?

```
def gen():
    yield 1
    yield 2
    return 3
g = gen()
print(next(g))
print(next(g))
```

A. 1 and 2
B. 1 and 3
C. 1, 2 and 3
D. Error

Answer: _____

Q16. What happens when the third next() is executed?

```
def numbers():
    yield 10
    yield 20
g = numbers()
next(g)
next(g)
print(next(g))
```

A. Prints None
B. Prints 20
C. Raises StopIteration
D. Restarts from 10

Answer: _____

Q17. What is the output?

```
x = 10
f = lambda y: x + y
x = 20
print(f(5))
```

A. 15
B. 25
C. 5
D. Error

Answer: _____

Q18. What is the result?

```
nums = [1, 2, 3, 4, 5]
result = list(map(lambda x: x * 2,
                  filter(lambda x: x % 2 == 1, nums)))
print(result)
```

A. [2, 4, 6, 8, 10]
B. [2, 6, 10]
C. [1, 3, 5]
D. [4, 8]

Answer: _____

Q19. What is printed?

```
from functools import reduce
nums = [2, 3, 4]
result = reduce(lambda a, b: a * b, nums, 2)
print(result)
```

A. 24
B. 48
C. 9
D. Error

Answer: _____

Q20. Which statement about map(), filter(), and generators is correct in Python 3?

A. map() and filter() always return lists
B. Generators can only be iterated using next()
C. map() and filter() return lazy iterators, and generator values are produced on demand
D. A generator automatically restarts after exhaustion

Answer: _____

Section B – Coding Questions

Q21. Find Prime Numbers in a Range

Write a Python program using nested loops and if conditions to print all prime numbers between 10 and 50.

Example input/output:

11 13 17 19 23 29 31 37 41 43 47

Q22. Count Vowels and Consonants

Write a function that accepts a string and returns the number of vowels and consonants. Ignore spaces, numbers, and special characters.

Example input/output:

Input: 'Hello Python'
Vowels: 3
Consonants: 8

Q23. Find the Second Largest Unique Number

Write a function that accepts a list of integers and returns the second largest distinct number without using sort() or sorted(). Assume at least two distinct values.

Example input/output:

Input: [10, 25, 15, 25, 8]
Output: 15

Q24. Create a Calculator Module and Package

Create a package named mymath containing operations.py with add(), subtract(), and multiply(). In main.py, import the functions using the package namespace and display results.

Example input/output:

add(10, 5) → 15
subtract(10, 5) → 5
multiply(10, 5) → 50

Q25. Build a Flexible Student Details Function

Write student_details(name, *marks, **details) accepting a name, subject marks and optional details such as city and grade. Print name, total marks, average marks and optional details. Assume at least one mark.

Example input/output:

student_details('Ravi', 80, 90, 70, city='Hyderabad', grade='A')
Total: 240
Average: 80.0
City: Hyderabad
Grade: A

Q26. Create a Generator for Even Numbers

Write even_numbers(n) yielding even numbers from 2 through n inclusive. Retrieve the first two values using next() and print remaining values with a for loop.

Example input/output:

Input: n = 10
First two: 2, 4
Remaining: 6, 8, 10

Q27. Process a List Using map(), filter(), and lambda

Given a list of integers, use filter() to select numbers greater than 10 and map() with lambda to calculate squares of selected numbers.

Example input/output:

Input: [5, 12, 8, 20, 15]
Output: [144, 400, 225]

Q28. Calculate Product Using reduce()

Write a program that uses reduce() with lambda to calculate the product of all positive numbers in a list. Assume at least one positive number.

Example input/output:

Input: [2, -3, 4, 0, 5]
Output: 40

Q29. Generate a Running Total

Write running_total(numbers) yielding the cumulative sum after processing each number in a list. Print generated values using a loop.

Example input/output:

Input: [5, 10, 3, 2]
Output: 5 15 18 20

Q30. Find Employees Eligible for a Salary Increment

Write a Python function eligible_employees(employees, min_experience=3) that accepts a dictionary containing employee names and years of experience.

Using filter(), map(), and lambda, select employees with at least 3 years of experience and calculate their updated experience after adding 1 year.